

Brutal Practice Paper B05 - Weather/Transportation/Data

Handling/Lines & Angles

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A maximum-minimum thermometer reads max 34 deg C and min 19 deg C on Monday, and max 30 deg C and min 21 deg C on Tuesday. By how much does Monday's temperature RANGE exceed Tuesday's range? [\[Weather, Climate & Adaptations\]](#)
 - (A) 6 deg C
 - (B) 4 deg C
 - (C) 9 deg C
 - (D) 7 deg C
 - (E) 15 deg C
 - (F) 2 deg C

2. The mean of 8 numbers is 15. If one of the numbers, 22, is removed, what is the mean of the remaining 7 numbers? [\[Data Handling\]](#)
 - (A) 17
 - (B) 14
 - (C) 12
 - (D) 13
 - (E) 16
 - (F) 98

3. The human heart beats about 72 times each minute at rest. About how many times does it beat in 2 hours? [\[Transportation in Animals & Plants\]](#)
 - (A) 144
 - (B) 4320
 - (C) 17280
 - (D) 720
 - (E) 8640
 - (F) 864
 - (G) 1440

4. Two angles are complementary, and one is 4 times the other. What is the LARGER angle? [\[Lines & Angles\]](#)
 - (A) 72 deg
 - (B) 36 deg
 - (C) 90 deg
 - (D) 18 deg
 - (E) 22.5 deg
 - (F) 54 deg
 - (G) 144 deg

5. Over 5 days the highest temperatures were 31, 35, 29, 34 and 36 deg C, and the lowest were 22, 24, 20, 23 and 21 deg C. What is the AVERAGE daily temperature range? [\[Weather, Climate & Adaptations\]](#)
- (A) 22 deg C
 - (B) 55 deg C
 - (C) 15 deg C
 - (D) 11 deg C
 - (E) 9 deg C
 - (F) 12 deg C
 - (G) 10 deg C
6. Find the median of the data: 12, 7, 15, 9, 20, 7, 11. [\[Data Handling\]](#)
- (A) 10
 - (B) 20
 - (C) 11
 - (D) 15
 - (E) 7
 - (F) 12
 - (G) 9
7. A nurse counts 18 pulse beats in 15 seconds. What is the pulse rate per minute? [\[Transportation in Animals & Plants\]](#)
- (A) 18
 - (B) 72
 - (C) 60
 - (D) 270
 - (E) 90
 - (F) 36
8. Two supplementary angles are in the ratio 2 : 7. What is the SMALLER angle? [\[Lines & Angles\]](#)
- (A) 40 deg
 - (B) 36 deg
 - (C) 140 deg
 - (D) 126 deg
 - (E) 20 deg
 - (F) 45 deg
9. In one day a town records 12 mm of rain. A flat rectangular terrace measures 6 m by 5 m. If 1 mm of rain on 1 square metre gives 1 litre, how much water falls on the terrace? [\[Weather, Climate & Adaptations\]](#)
- (A) 180 litres
 - (B) 72 litres
 - (C) 60 litres
 - (D) 720 litres
 - (E) 30 litres
 - (F) 360 litres
 - (G) 36 litres

10. Find the mode of: 4, 6, 4, 8, 6, 4, 9, 6, 4. [\[Data Handling\]](#)

- (A) 8
- (B) 9
- (C) 4
- (D) 6
- (E) 5
- (F) 47
- (G) 7

11. At rest the heart beats 72 times a minute; during exercise it beats 120 times a minute. How many EXTRA beats occur over 5 minutes of exercise compared with resting? [\[Transportation in Animals & Plants\]](#)

- (A) 360
- (B) 240
- (C) 120
- (D) 48
- (E) 600
- (F) 192
- (G) 300

12. An angle is 30 deg more than its own complement. What is the angle? [\[Lines & Angles\]](#)

- (A) 45 deg
- (B) 30 deg
- (C) 55 deg
- (D) 60 deg
- (E) 75 deg
- (F) 15 deg

13. Which statement describes the CLIMATE of a place rather than its weather? [\[Weather, Climate & Adaptations\]](#)

- (A) It snowed in Shimla today
- (B) Today is hotter than yesterday
- (C) There was a thunderstorm last night
- (D) The sky is cloudy this morning
- (E) Cherrapunji receives very heavy rainfall every year for many years
- (F) It is raining in Delhi this afternoon

14. A bag has 6 red, 3 blue and 1 green ball. One ball is drawn at random. What is the probability that it is NOT red? [\[Data Handling\]](#)

- (A) $\frac{4}{5}$
- (B) $\frac{6}{10}$
- (C) $\frac{2}{5}$
- (D) $\frac{1}{10}$
- (E) $\frac{3}{10}$
- (F) $\frac{3}{5}$
- (G) $\frac{1}{2}$

15. Which part of blood mainly helps the body fight germs and disease? [\[Transportation in Animals & Plants\]](#)

- (A) Platelets
- (B) White blood cells
- (C) Haemoglobin
- (D) Serum only
- (E) Red blood cells
- (F) Water

16. Two angles $(2x + 10)$ deg and $(3x + 20)$ deg form a linear pair. Find the value of x . [\[Lines & Angles\]](#)

- (A) 150
- (B) 34
- (C) 25
- (D) 30
- (E) 60
- (F) 36

17. At 6 a.m. the temperature is 18 deg C. It rises 3 deg C each hour until 12 noon, then falls 2 deg C each hour until 4 p.m. What is the temperature at 4 p.m.? [\[Weather, Climate & Adaptations\]](#)

- (A) 36 deg C
- (B) 12 deg C
- (C) 34 deg C
- (D) 28 deg C
- (E) 26 deg C
- (F) 30 deg C

18. The average of 5 numbers is 18. Four of them are 14, 20, 25 and 11. Find the fifth number. [\[Data Handling\]](#)

- (A) 16
- (B) 22
- (C) 24
- (D) 90
- (E) 70
- (F) 20
- (G) 18

19. A plant loses about 250 mL of water by transpiration each day. How much water (in litres) does it lose during the whole month of June (30 days)? [\[Transportation in Animals & Plants\]](#)

- (A) 0.075 litres
- (B) 7.2 litres
- (C) 0.75 litres
- (D) 75 litres
- (E) 7500 litres
- (F) 7.5 litres
- (G) 750 litres

20. Two straight lines cross. One of the four angles is 65 deg. What is the measure of an angle ADJACENT to it (forming a straight line with it)? [\[Lines & Angles\]](#)

- (A) 65 deg
- (B) 25 deg
- (C) 180 deg
- (D) 35 deg
- (E) 115 deg
- (F) 130 deg
- (G) 295 deg

21. The mean of the maximum temperatures over 6 days is 33 deg C. On five of the days the maxima were 31, 35, 30, 34 and 33 deg C. Find the sixth day's maximum. [\[Weather, Climate & Adaptations\]](#)

- (A) 33 deg C
- (B) 35 deg C
- (C) 163 deg C
- (D) 36 deg C
- (E) 198 deg C
- (F) 32 deg C
- (G) 30 deg C

22. The marks scored by six students are 45, 78, 52, 90, 33 and 67. What is the range of the marks? [\[Data Handling\]](#)

- (A) 57
- (B) 45
- (C) 123
- (D) 47
- (E) 67
- (F) 53
- (G) 90

23. In the human heart, blood vessels that carry blood AWAY from the heart to the body are called what? [\[Transportation in Animals & Plants\]](#)

- (A) Veins
- (B) Arteries
- (C) Valves
- (D) Nerves
- (E) Lymph tubes
- (F) Capillaries
- (G) Chambers

24. A transversal cuts two parallel lines. One pair of co-interior (same-side interior) angles has one angle equal to 70 deg. What is the other co-interior angle? [\[Lines & Angles\]](#)

- (A) 110 deg
- (B) 20 deg
- (C) 70 deg
- (D) 160 deg
- (E) 290 deg
- (F) 140 deg
- (G) 35 deg

25. A rain gauge records daily rainfall of 0, 14, 7, 0, 21, 5 and 9 mm in a week. What is the mean daily rainfall for the week? [\[Weather, Climate & Adaptations\]](#)

- (A) 8 mm
- (B) 56 mm
- (C) 7 mm
- (D) 9 mm
- (E) 10 mm
- (F) 11 mm
- (G) 21 mm

26. In four sections A, B, C and D, the number of pupils who like Maths is 20, 15, 25 and 30; the number who like Science is 18, 22, 20 and 25. In how many sections do MORE pupils like Science than Maths? [\[Data Handling\]](#)

- (A) 4
- (B) 0
- (C) 5
- (D) 3
- (E) 1
- (F) 2
- (G) 8

27. Water and dissolved minerals are carried UPWARD from the roots to the leaves through which plant tissue? [\[Transportation in Animals & Plants\]](#)

- (A) Xylem
- (B) Veins of the petal
- (C) Stomata
- (D) Root hairs only
- (E) Pith
- (F) Phloem
- (G) Bark

28. Two parallel lines are cut by a transversal. A corresponding angle is written as $3x$ and the matching given angle is 120° . Find x . [\[Lines & Angles\]](#)

- (A) 30
- (B) 20
- (C) 40
- (D) 45
- (E) 360
- (F) 60
- (G) 120

29. Which of these is NOT an adaptation that helps an animal survive in a hot desert? [\[Weather, Climate & Adaptations\]](#)

- (A) Long eyelashes to keep out blowing sand
- (B) Looking for food mainly at night
- (C) Being able to drink a large amount of water at once
- (D) A thick layer of blubber (fat) under the skin
- (E) Tolerating a high body temperature
- (F) Storing fat in a hump

30. The mean of five consecutive whole numbers is 23. What is the largest of these five numbers? [\[Data](#)

[Handling\]](#)

- (A) 27
- (B) 25
- (C) 115
- (D) 26
- (E) 24
- (F) 21

31. A child's heart beats 90 times each minute. How many times does it beat in a full day (24 hours)?

[\[Transportation in Animals & Plants\]](#)

- (A) 21600
- (B) 64800
- (C) 1296000
- (D) 2160
- (E) 12960
- (F) 129600
- (G) 5400

32. Two parallel lines are cut by a transversal. The alternate interior angles are $(5x - 10)$ deg and $(3x + 30)$ deg. What is the measure of each angle? [\[Lines & Angles\]](#)

- (A) 80 deg
- (B) 100 deg
- (C) 30 deg
- (D) 90 deg
- (E) 20 deg
- (F) 45 deg
- (G) 60 deg

33. A migratory flock flies 1200 km in 3 days, flying 8 hours on each of the days. What is its average flying speed? [\[Weather, Climate & Adaptations\]](#)

- (A) 60 km/h
- (B) 100 km/h
- (C) 25 km/h
- (D) 400 km/h
- (E) 48 km/h
- (F) 150 km/h
- (G) 50 km/h

34. A fair die is rolled once. What is the probability of getting a number that is BOTH even AND greater than 3? [\[Data Handling\]](#)

- (A) $\frac{1}{4}$
- (B) $\frac{1}{6}$
- (C) $\frac{1}{3}$
- (D) $\frac{1}{2}$
- (E) $\frac{5}{6}$
- (F) $\frac{1}{5}$
- (G) $\frac{2}{3}$

35. A nurse records a patient's pulse on five visits as 76, 80, 78, 82 and 84 beats per minute. What is the mean pulse? [\[Transportation in Animals & Plants\]](#)

- (A) 78
- (B) 80
- (C) 76
- (D) 400
- (E) 82
- (F) 84
- (G) 79

36. Three angles lie on a straight line and are x , $2x$ and $3x$. What is the LARGEST of the three angles?

[\[Lines & Angles\]](#)

- (A) 120 deg
- (B) 60 deg
- (C) 90 deg
- (D) 45 deg
- (E) 180 deg
- (F) 100 deg
- (G) 30 deg

37. In June a city's average maximum is 38 deg C and average minimum is 26 deg C. A heat wave raises every maximum by 4 deg C and every minimum by 1 deg C. What is the new average DAILY RANGE?

[\[Weather, Climate & Adaptations\]](#)

- (A) 9 deg C
- (B) 5 deg C
- (C) 12 deg C
- (D) 13 deg C
- (E) 15 deg C
- (F) 16 deg C
- (G) 14 deg C

38. The mean weight of 10 boys is 35 kg and the mean weight of 15 girls is 30 kg. What is the mean weight of all 25 children together? [\[Data Handling\]](#)

- (A) 32 kg
- (B) 65 kg
- (C) 35 kg
- (D) 33 kg
- (E) 30 kg
- (F) 800 kg

39. Which of the following is NOT a waste product that the body removes (an excretory product)?

[\[Transportation in Animals & Plants\]](#)

- (A) Urine
- (B) Glucose (digested food)
- (C) Excess water
- (D) Urea
- (E) Sweat
- (F) Carbon dioxide
- (G) Extra salts

40. Four angles meet at a point. Three of them are 90 deg, 85 deg and 75 deg. What is the fourth angle?

[Lines & Angles]

- (A) 250 deg
- (B) 100 deg
- (C) 70 deg
- (D) 105 deg
- (E) 110 deg
- (F) 90 deg
- (G) 120 deg

41. Town A received 45 mm of rain over 5 days and Town B received 72 mm over 8 days. Which town had the higher AVERAGE daily rainfall? [Weather, Climate & Adaptations]

- (A) Cannot be compared
- (B) Town B, by 9 mm per day
- (C) Town A, by 3 mm per day
- (D) Town B, by 27 mm per day
- (E) Town A, by 27 mm per day
- (F) Both are equal at 9 mm per day

42. Find the median of: 8, 12, 5, 20, 15, 9. [Data Handling]

- (A) 12
- (B) 13.5
- (C) 10.5
- (D) 10
- (E) 9
- (F) 8
- (G) 11

43. A leaf loses about 0.5 mL of water per square cm of surface each day. A leaf has an area of 24 square cm. How much water does it transpire in 3 days? [Transportation in Animals & Plants]

- (A) 12 mL
- (B) 24 mL
- (C) 72 mL
- (D) 8 mL
- (E) 48 mL
- (F) 36 mL
- (G) 18 mL

44. Find the DIFFERENCE between the supplement and the complement of a 40 deg angle. [Lines & Angles]

- (A) 70 deg
- (B) 180 deg
- (C) 100 deg
- (D) 80 deg
- (E) 50 deg
- (F) 90 deg

45. In a tropical rainforest, many leaves have pointed 'drip-tips'. What is the main use of a drip-tip? [\[Weather, Climate & Adaptations\]](#)
- (A) It keeps the leaf warm at night
 - (B) It lets heavy rainwater run off the leaf quickly
 - (C) It catches sunlight more strongly
 - (D) It reflects rain away from the roots
 - (E) It stores water for the dry season
 - (F) It helps the leaf float on water
46. For the data 3, 3, 3, 9, 12, which is greater - the mean or the mode - and by how much? [\[Data Handling\]](#)
- (A) They are equal
 - (B) The mode is greater, by 6
 - (C) The mean is greater, by 9
 - (D) The mean is greater, by 3
 - (E) The mean is greater, by 6
 - (F) The mode is greater, by 9
 - (G) The mode is greater, by 3
47. After sprinting, a boy's heart beats 150 times a minute. It slows by 12 beats each minute of rest until it reaches his resting rate of 78. After how many minutes of rest does it reach 78? [\[Transportation in Animals & Plants\]](#)
- (A) 72 minutes
 - (B) 9 minutes
 - (C) 6 minutes
 - (D) 5 minutes
 - (E) 7 minutes
 - (F) 8 minutes
 - (G) 12 minutes
48. The supplement of an angle is 4 times the angle itself. What is the angle? [\[Lines & Angles\]](#)
- (A) 40 deg
 - (B) 36 deg
 - (C) 45 deg
 - (D) 144 deg
 - (E) 135 deg
 - (F) 72 deg
 - (G) 30 deg
49. The mean of the maximum temperatures for a 7-day week is 32 deg C. An eighth day with a maximum of 40 deg C is then added. What is the new mean of the 8 days? [\[Weather, Climate & Adaptations\]](#)
- (A) 32 deg C
 - (B) 35 deg C
 - (C) 36 deg C
 - (D) 264 deg C
 - (E) 34 deg C
 - (F) 40 deg C
 - (G) 33 deg C

50. The mean of 6 numbers is 40. If every number is increased by 5, what is the new mean? [\[Data Handling\]](#)

- (A) 45
- (B) 240
- (C) 35
- (D) 41
- (E) 50
- (F) 46

51. Transpiration is the loss of water as vapour from a plant. Besides removing extra water, what useful effect does transpiration have? [\[Transportation in Animals & Plants\]](#)

- (A) It traps insects
- (B) It thickens the bark
- (C) It helps pull water and minerals up the plant and cools it
- (D) It colours the flowers
- (E) It makes food for the plant
- (F) It stores starch in the leaves
- (G) It lets the plant breathe out oxygen at night

52. Two complementary angles differ by 16 deg. What is the LARGER angle? [\[Lines & Angles\]](#)

- (A) 74 deg
- (B) 37 deg
- (C) 32 deg
- (D) 53 deg
- (E) 61 deg
- (F) 45 deg
- (G) 58 deg

53. On one day a desert town cools from 45 deg C at noon to 12 deg C at midnight, while a mountain town cools from 20 deg C to -5 deg C. Which town has the LARGER daily temperature variation, and by how much? [\[Weather, Climate & Adaptations\]](#)

- (A) Desert town, by 5 deg C
- (B) Desert town, by 33 deg C
- (C) Desert town, by 8 deg C
- (D) Mountain town, by 8 deg C
- (E) Mountain town, by 25 deg C
- (F) They are equal

54. A spinner has 8 equal sectors numbered 1 to 8. What is the probability that the arrow stops on a PRIME number? [\[Data Handling\]](#)

- (A) $\frac{1}{2}$
- (B) $\frac{3}{4}$
- (C) $\frac{5}{8}$
- (D) $\frac{1}{8}$
- (E) $\frac{2}{8}$
- (F) $\frac{1}{4}$
- (G) $\frac{3}{8}$

55. The red colour of blood and its ability to carry oxygen are both due to which substance in red blood cells? [\[Transportation in Animals & Plants\]](#)

- (A) Chlorophyll
- (B) Haemoglobin
- (C) Plasma
- (D) White cells
- (E) Water
- (F) Serum

56. Two angles of a linear pair are in the ratio 4 : 5. What is the BIGGER angle? [\[Lines & Angles\]](#)

- (A) 120 deg
- (B) 100 deg
- (C) 20 deg
- (D) 80 deg
- (E) 45 deg
- (F) 36 deg

57. A rain gauge has a circular mouth of area 200 square cm. After a storm it holds 600 cubic cm (mL) of water. What depth of rainfall does this represent? [\[Weather, Climate & Adaptations\]](#)

- (A) 0.3 mm
- (B) 30 mm
- (C) 300 mm
- (D) 60 mm
- (E) 120 mm
- (F) 3 mm

58. The mean of 7 observations is 12. Six of them are 9, 15, 11, 13, 8 and 14. Find the seventh observation. [\[Data Handling\]](#)

- (A) 16
- (B) 12
- (C) 70
- (D) 14
- (E) 10
- (F) 13

59. An athlete's resting pulse is 50 beats a minute, while a typical adult's is 72. Over one hour, how many FEWER beats does the athlete's heart make than the typical adult's? [\[Transportation in Animals & Plants\]](#)

- (A) 3000
- (B) 132
- (C) 22
- (D) 1100
- (E) 4320
- (F) 660
- (G) 1320

60. Two parallel lines are cut by a transversal. The co-interior angles are $(2x + 15)$ deg and $(3x + 25)$ deg.

Find x . [\[Lines & Angles\]](#)

- (A) 40
- (B) 56
- (C) 140
- (D) 35
- (E) 28
- (F) 25

61. Which kind of place usually has the SMALLEST difference between its summer and winter temperatures? [\[Weather, Climate & Adaptations\]](#)

- (A) A place close to the poles
- (B) A place close to the equator
- (C) A snowy mountain peak
- (D) A high mountain in a cold country
- (E) A polar ice cap
- (F) A place in the far north of Russia
- (G) A desert far inland

62. Over 4 weeks the rainfall was 20, 35, 15 and 30 mm. What was the mean weekly rainfall? [\[Data Handling\]](#)

- (A) 35 mm
- (B) 24 mm
- (C) 100 mm
- (D) 25 mm
- (E) 20 mm
- (F) 30 mm
- (G) 50 mm

63. The pale yellowish liquid part of blood, in which the cells float, is called what? [\[Transportation in Animals & Plants\]](#)

- (A) Platelets
- (B) Plasma
- (C) Lymph nodes
- (D) Haemoglobin
- (E) Urine
- (F) Bile
- (G) Marrow

64. An angle of 130 deg is bisected, and then one of the halves is bisected again. What is the SMALLEST angle produced? [\[Lines & Angles\]](#)

- (A) 45 deg
- (B) 32.5 deg
- (C) 65 deg
- (D) 26 deg
- (E) 16.25 deg
- (F) 130 deg
- (G) 43.33 deg

65. A migratory bird must cover 6400 km. It flies 400 km on day 1 and 100 km more each following day (500 on day 2, and so on). On which day does it first fly 800 km? [\[Weather, Climate & Adaptations\]](#)

- (A) Day 2
- (B) Day 7
- (C) Day 5
- (D) Day 6
- (E) Day 3
- (F) Day 8
- (G) Day 4

66. A bag contains only black balls. What is the probability of drawing a white ball from it? [\[Data Handling\]](#)

- (A) 1
- (B) $\frac{1}{10}$
- (C) $\frac{1}{100}$
- (D) $\frac{1}{2}$
- (E) 0
- (F) Cannot say
- (G) $\frac{1}{8}$

67. Tiny pores on the underside of a leaf, through which water vapour escapes during transpiration, are called what? [\[Transportation in Animals & Plants\]](#)

- (A) Petals
- (B) Xylem tubes
- (C) Lenticels
- (D) Stomata
- (E) Phloem
- (F) Veins

68. What is the REFLEX angle that corresponds to an angle of 110 deg? [\[Lines & Angles\]](#)

- (A) 360 deg
- (B) 290 deg
- (C) 250 deg
- (D) 220 deg
- (E) 110 deg
- (F) 180 deg
- (G) 70 deg

69. A weather diary shows these daily highs for a week: 30, 33, 30, 31, 30, 34 and 30 deg C. Which value is the MODE, and is it higher or lower than the mean? [\[Weather, Climate & Adaptations\]](#)

- (A) Mode 30 deg C, higher than the mean
- (B) Mode 34 deg C, higher than the mean
- (C) Mode 30 deg C, equal to the mean
- (D) There is no mode
- (E) Mode 31 deg C, equal to the mean
- (F) Mode 30 deg C, which is lower than the mean

70. A student's mean across 5 tests was 76. Later it is found that a test recorded as 60 was actually 90. What is the corrected mean? [\[Data Handling\]](#)

- (A) 78
- (B) 82
- (C) 86
- (D) 90
- (E) 84
- (F) 80
- (G) 76

71. A doctor counts 26 heartbeats in 20 seconds. What is the heart rate per minute? [\[Transportation in Animals & Plants\]](#)

- (A) 104
- (B) 84
- (C) 78
- (D) 156
- (E) 52
- (F) 39
- (G) 26

72. Two adjacent angles lie on a straight line. One angle is 30 deg more than twice the other. What is the LARGER angle? [\[Lines & Angles\]](#)

- (A) 50 deg
- (B) 130 deg
- (C) 100 deg
- (D) 120 deg
- (E) 150 deg
- (F) 110 deg
- (G) 60 deg

73. Polar bears have a thick layer of fat and dense fur. A bear's fur reduces its heat loss to one-quarter of what bare skin would lose. If bare skin would lose 240 units of heat per hour, how much does the furred bear lose? [\[Weather, Climate & Adaptations\]](#)

- (A) 120 units per hour
- (B) 960 units per hour
- (C) 180 units per hour
- (D) 30 units per hour
- (E) 236 units per hour
- (F) 80 units per hour
- (G) 60 units per hour

- 74.** In a data set the maximum is 60 and the minimum is 12, giving a range of 48. If the maximum rises by 5 and the minimum falls by 3, what is the new range? [\[Data Handling\]](#)
- (A) 48
 - (B) 53
 - (C) 8
 - (D) 60
 - (E) 51
 - (F) 45
 - (G) 56
- 75.** A large tree draws up about 40 litres of water a day. About 99% is lost by transpiration and the rest is used by the tree. How many litres does the tree actually USE in a day? [\[Transportation in Animals & Plants\]](#)
- (A) 1 litre
 - (B) 0.4 litres
 - (C) 4 litres
 - (D) 0.04 litres
 - (E) 39.6 litres
 - (F) 40 litres
- 76.** Two parallel lines are cut by a transversal so that the two co-interior angles are in the ratio 1 : 3. What is the SMALLER co-interior angle? [\[Lines & Angles\]](#)
- (A) 36 deg
 - (B) 60 deg
 - (C) 45 deg
 - (D) 40 deg
 - (E) 135 deg
 - (F) 30 deg
- 77.** Over four months a hill station's average rainfall was 120, 150, 90 and 60 mm. What was the mean MONTHLY rainfall, and which month was farthest from this mean? [\[Weather, Climate & Adaptations\]](#)
- (A) Mean 105 mm; the 60 mm month is farthest
 - (B) Mean 120 mm; the 60 mm month is farthest
 - (C) Mean 105 mm; the 90 mm month is farthest
 - (D) Mean 105 mm; the 120 mm month is farthest
 - (E) Mean 105 mm; the 150 mm month is farthest
 - (F) Mean 100 mm; the 90 mm month is farthest
- 78.** Cards numbered 1 to 20 are placed in a box and one is drawn at random. What is the probability of drawing a multiple of 4? [\[Data Handling\]](#)
- (A) $\frac{2}{5}$
 - (B) $\frac{3}{20}$
 - (C) $\frac{1}{2}$
 - (D) $\frac{1}{20}$
 - (E) $\frac{1}{4}$
 - (F) $\frac{1}{10}$
 - (G) $\frac{1}{5}$

79. Veins contain valves along their length. What is the main job of these valves? [\[Transportation in Animals & Plants\]](#)

- (A) To make new blood cells
- (B) To cool the blood
- (C) To stop blood from flowing backward
- (D) To speed the blood up
- (E) To filter out germs
- (F) To add oxygen to the blood
- (G) To store extra blood

80. The complement of an angle is taken; then the SUPPLEMENT of that result is found. If the original angle is 25 deg, what is the final angle? [\[Lines & Angles\]](#)

- (A) 90 deg
- (B) 75 deg
- (C) 65 deg
- (D) 155 deg
- (E) 140 deg
- (F) 115 deg
- (G) 25 deg

81. A camel can go many days without drinking, then drink about 100 litres in 10 minutes. At that steady rate, how much would it drink in 4 minutes? [\[Weather, Climate & Adaptations\]](#)

- (A) 10 litres
- (B) 40 litres
- (C) 250 litres
- (D) 25 litres
- (E) 50 litres
- (F) 400 litres
- (G) 4 litres

82. The mean age of a family of 6 is 25 years. The grandfather, aged 60, moves away. What is the mean age of the remaining 5 members? [\[Data Handling\]](#)

- (A) 18 years
- (B) 22 years
- (C) 15 years
- (D) 30 years
- (E) 90 years
- (F) 25 years

83. The actual exchange of oxygen, food and wastes between blood and the body's cells happens in which vessels? [\[Transportation in Animals & Plants\]](#)

- (A) The heart chambers
- (B) Arteries
- (C) Capillaries
- (D) Valves
- (E) Xylem
- (F) The aorta
- (G) Large veins

84. Four angles around a point are in the ratio 1 : 2 : 3 : 4. What is the LARGEST angle? [\[Lines & Angles\]](#)

- (A) 144 deg
- (B) 150 deg
- (C) 72 deg
- (D) 120 deg
- (E) 108 deg
- (F) 36 deg
- (G) 90 deg

85. Snow on a mountain reflects most sunlight, so the air stays cold. If fresh snow reflects 80% of incoming sunlight and 500 units of sunlight arrive, how many units are reflected? [\[Weather, Climate & Adaptations\]](#)

- (A) 300 units
- (B) 420 units
- (C) 250 units
- (D) 480 units
- (E) 100 units
- (F) 400 units

86. A class of 40 pupils scored a mean of 65, and another class of 10 pupils scored a mean of 90 on the same test. What is the combined mean of all 50 pupils? [\[Data Handling\]](#)

- (A) 70
- (B) 65
- (C) 80
- (D) 75
- (E) 72
- (F) 3500
- (G) 77.5

87. The heart pumps about 70 mL of blood with each beat. Beating 72 times a minute, about how many litres does it pump in one minute? [\[Transportation in Animals & Plants\]](#)

- (A) 5.04 litres
- (B) 504 litres
- (C) 0.0504 litres
- (D) 5.4 litres
- (E) 5040 litres
- (F) 0.504 litres
- (G) 50.4 litres

88. Two lines intersect. One of the angles is 75 deg. The angle ADJACENT to it (its linear-pair partner) is then bisected. What is the measure of each half? [\[Lines & Angles\]](#)

- (A) 75 deg
- (B) 37.5 deg
- (C) 52.5 deg
- (D) 32.5 deg
- (E) 60 deg
- (F) 90 deg
- (G) 105 deg

89. A maximum-minimum thermometer over 30 days gives a mean maximum of 36 deg C and a mean minimum of 24 deg C. What is the average mid-value temperature, taken as halfway between the daily max and min? [\[Weather, Climate & Adaptations\]](#)

- (A) 6 deg C
- (B) 48 deg C
- (C) 12 deg C
- (D) 27 deg C
- (E) 60 deg C
- (F) 30 deg C
- (G) 33 deg C

90. For the data 2, 4, 4, 6, 6, 8, which statement about the mode is correct? [\[Data Handling\]](#)

- (A) There are two modes, 4 and 6
- (B) The mode is 4
- (C) There is no mode
- (D) The mode is 5
- (E) The mode is 6
- (F) The mode is 8
- (G) The mode is 2

91. How does a plant get rid of most of its extra water? [\[Transportation in Animals & Plants\]](#)

- (A) By pushing it back into the soil through roots
- (B) By turning it into food
- (C) Through the flowers as nectar
- (D) As water vapour through the stomata of its leaves
- (E) Through tiny mouths that swallow it
- (F) By sweating like animals
- (G) By storing it forever in the stem

92. For an angle of 35 deg, find the SUM of its complement and its supplement. [\[Lines & Angles\]](#)

- (A) 290 deg
- (B) 110 deg
- (C) 90 deg
- (D) 235 deg
- (E) 180 deg
- (F) 200 deg
- (G) 160 deg

93. A desert plant (like a cactus) is adapted to its climate. Which feature is its adaptation to save water?

[\[Weather, Climate & Adaptations\]](#)

- (A) Leaves reduced to spines to cut water loss
- (B) Bright wide petals open all day
- (C) Broad flat leaves to catch more rain
- (D) Many large pores on every leaf
- (E) Shallow leaves that float
- (F) Long soft drip-tips
- (G) A thin papery bark

94. A month of the year is chosen at random. What is the probability that its name begins with the letter 'J'?

[Data Handling]

- (A) $\frac{1}{4}$
- (B) $\frac{3}{4}$
- (C) $\frac{1}{2}$
- (D) $\frac{1}{6}$
- (E) $\frac{1}{3}$
- (F) $\frac{2}{12}$
- (G) $\frac{1}{12}$

95. Sweating helps cool the body. A worker loses 1.5 litres of sweat in 3 hours of steady work. At the same rate, how much sweat is lost in 8 hours? [Transportation in Animals & Plants]

- (A) 2 litres
- (B) 12 litres
- (C) 3 litres
- (D) 4.5 litres
- (E) 6 litres
- (F) 0.5 litres
- (G) 4 litres

96. Two parallel lines are cut by a transversal. A pair of alternate interior angles is $(4x)$ deg and $(x + 60)$ deg. What is the measure of each angle? [Lines & Angles]

- (A) 100 deg
- (B) 20 deg
- (C) 80 deg
- (D) 120 deg
- (E) 40 deg
- (F) 60 deg

97. The average temperature for the first 4 days of a week is 28 deg C, and for the last 3 days is 35 deg C. What is the mean temperature for the whole 7-day week? [Weather, Climate & Adaptations]

- (A) 33 deg C
- (B) 63 deg C
- (C) 31 deg C
- (D) 31.5 deg C
- (E) 32 deg C
- (F) 29 deg C
- (G) 30 deg C

98. Find the mean of the decimals 12.5, 13.5, 14 and 10. [Data Handling]

- (A) 12
- (B) 12.5
- (C) 50
- (D) 11.5
- (E) 12.75
- (F) 13

99. A patient receives fluid through a drip at 25 drops per minute, where 20 drops make 1 mL. How many mL of fluid enter the body in 2 hours? [\[Transportation in Animals & Plants\]](#)

- (A) 75 mL
- (B) 150 mL
- (C) 3000 mL
- (D) 1.25 mL
- (E) 120 mL
- (F) 300 mL
- (G) 50 mL

100. An angle is exactly equal to its own supplement. What is the angle? [\[Lines & Angles\]](#)

- (A) 120 deg
- (B) 60 deg
- (C) 180 deg
- (D) 100 deg
- (E) 30 deg
- (F) 45 deg
- (G) 90 deg